Alok Gaur

M.Tech Computational Mathematics
Jamia Millia Islamia University

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Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2026 – 2028	M.Tech (Computational Mathematics)	Jamia Millia Islamia University	–
2026	B.Tech (Computer Science Eng.)	MERI CET, Delhi-NCR	77.4
2022	High School Diploma	GBSSS No.3, Najafgarh, New Delhi	85.4

Summary

M.Tech Computational Mathematics student with a Computer Science background and hands-on experience in AI/ML, Generative AI, NLP, and Computer Vision. Proficient in Python, PyTorch, TensorFlow, FastAPI, and LLM-based systems, with experience building AI agents, RAG applications, and deep-learning solutions.

Work Experience

DIGISAMAKSH Private Limited — Data Science Intern Jun'25 – Aug'25

- Assisted in collecting, cleaning, and analyzing large datasets to derive data-driven business insights.
- Conducted exploratory data analysis (EDA) and applied statistical techniques to optimize business operations.
- Developed and trained predictive models and neural networks to improve decision-making accuracy.

Ibest Multiple Work Private Limited — Python Intern Jan'24 – Jul'24

- Engineered high-performance backend APIs using FastAPI, achieving rapid response times under 100ms.
- Integrated internal services with government and non-government portals, enhancing data accessibility and reducing manual entry workloads by 10%.

Key Projects

Mail Management: Autonomous AI Agent System *[Ongoing / Solo]*

- **Tech Stack:** Python, FastAPI, Google Mail APIs, Ollama Qwen3, Smolagents, LiteLLM, PostgreSQL, ChromaDB, Redis
- Architected an agentic workflow that monitors user inboxes and intelligently classifies incoming mail including expenses, queries, and recruitment.
- Automated financial auditing logs, customer replies, and calendar scheduling using LLM-powered agents.

Talk With Document: RAG-Powered PDF Explorer *[Completed / Solo]*

- **Tech Stack:** Python, Ollama Qwen3, pdfplumber, Transformers, Sentence Transformers, MongoDB
- Developed a RAG-based AI system for real-time, context-aware querying and information extraction from lengthy documents.
- Streamlined document workflows by synthesizing key insights, summaries, and answering user prompts.

Conditional Generative Adversarial Network (CGAN) *[Completed / Solo]*

- **Tech Stack:** PyTorch, Google Colab, Kaggle Data Ecosystem
- Built and trained a CGAN for targeted image generation using precise input labels and a Kaggle bird dataset.
- Resolved model-collapse challenges and stabilized adversarial training for balanced image generation.

Image Restoration using Autoencoders *[Completed / Solo]*

- **Tech Stack:** TensorFlow, Keras, OpenCV, Python, Kaggle
- Designed a deep-learning restoration network using symmetric skip connections for enhanced image reconstruction.
- Engineered a pipeline for preprocessing, noise extraction, and high-fidelity output reconstruction.

Technical Skills

Languages: Python, SQL, JavaScript, HTML/CSS
AI/ML: PyTorch, TensorFlow, Keras, Hugging Face, LangChain, Scikit-Learn, OpenCV, NumPy, Pandas
Backend & Databases: FastAPI, PostgreSQL, MongoDB, Redis, ChromaDB
Cloud: AWS, GCP
AWS: EC2, S3, RDS, DynamoDB, ELB, IAM, VPC, NACL, CloudWatch, CloudFormation, Auto Scaling
Tools & Concepts: Git, GitHub, Ollama, Google Colab, Computer Vision, NLP, RAG, LLM Agents, Generative AI, Deep Learning

Achievements & Certifications

AWS Certified Cloud Practitioner
Foundation: Introduction to LangChain - Python Certificate by LangChain Academy
GATE Qualified (2026): Secured an All India Rank of 7812 in the Graduate Aptitude Test in Engineering.